

* why need the timer ?

↳ waiting time

① Block wait → -delay.ms

② timer

↳ Interrupt Trigger type

① Event trigger

② time trigger

$T_1 \rightarrow 50 \text{ msec}$ $T_2 \rightarrow 100 \text{ msec}$

$T_3 \rightarrow 200 \text{ msec}$

* timer / Counter

① Count Pulse (clock)

② from Pulse can get time

Processor time unit : * clock
human time unit : * sec.



③ and signal have frequency Parameter

↳ signal → $f_{req} \rightarrow 8 \text{ MHz}$

$$\text{Periodic time} = \frac{1}{f_{req}} = \frac{1}{8} = 125 \text{ } \mu\text{sec}$$

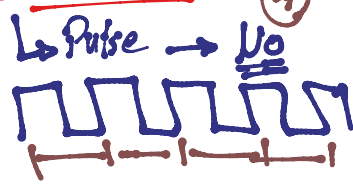
$$\text{clk time} = \frac{1}{f_{req}}$$

* why called timer / Counter

① timer

↳ Pulse → time

② Counter



* timer / Counter

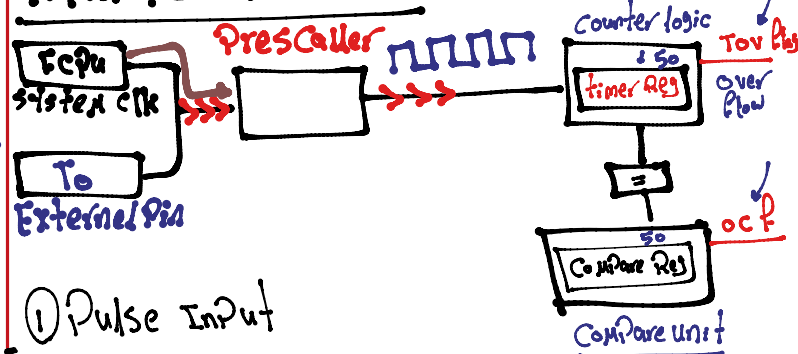
* Note in Atmega32

① we have 3 Peripheral timer.

↳ timer0 → 8 bit → size Reg
↳ timer1 → 16 bit →
↳ timer2 → 8 bit.

② we will explain the timer0 / 2 timer1 → timer0 + Advanced option.

timer / Counter H.W



① Pulse Input

② Prescaler

- ① Enable / disable
- ② select the pulse input
 - ↳ system Pres
 - ↳ External
- ③ if system Pres
 - ↳ select Factor No.
- ④ if External
 - ↳ select the pulse time.



Rising



Falling

③ Counter logic

- ① must have Register memory to store the count.
- ② any Register have limit (H.W limitation)
 - ↳ timer0 → timer Reg → size = 8 bit
 - ↳ $2^0 \rightarrow 2^8 \rightarrow 255$
 - ↳ 8 bit timer → 255
- ③ have the overflow flag

timer

* timer mode

① Normal Mode → [overflow ×]

↳ Flag TOV & OC are set.

② Compare match mode

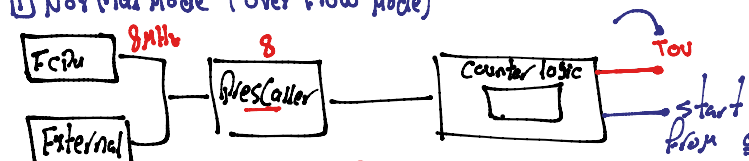
↳ clear timer compare (CTC) ×

↳ Flag OC

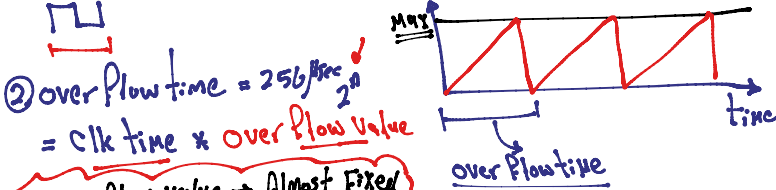
↳ PWM

③ Watchdog

① Normal Mode (overflow mode)



① clk time = $\frac{\text{Prescaler}}{\text{system freq.}} = \frac{8}{8\text{M}} = 1\text{ }\mu\text{sec.}$



② overflow time = $256 \times \text{clk time}$
= clk time × overflow value

* overflow value → Almost Fixed
No
* in another option can be variable

* Relation between Req time & ovf time.

- ① Req time = ovf time
- ② Req time > ovf time → 5 msec > 256 μsec
- ③ Req time < ovf time

* Req time > ovf time → need multi ovf to achieve Req time

③ # No of ovf Count = $\frac{\text{Req time}}{\text{ovf time}}$

↳ must be have same unit
↳ handle the count no if int / float.

No of ovf Count is int

void to_init() {
 Normal mode
 timer Reg = 0;
 Enable ovf interrupt
}

ISR(ovf) {
 static uint8_t Count = 0;
 Count++;
 if (Count == No of ovf)
 {
 Action; // scheduler
 Count = 0;
 }
}

No of ovf Count is float



Preload → start point
* Preload = overflow value × (1 - 0.4)
= 256 × (1 - 0.4)

* No of ovf Count is float

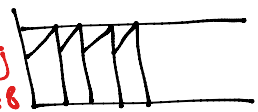
void to_init() {

Normal mode
 timer Register = preload total value.
 Enable ovf
}

ISR(ovf) {
 static uint8_t Count = 0;
 Count++;
 if (Count == No of ovf + 1)
 {
 Action ←
 Count = 0;
 timer Reg = preload;
 }
}

* Req time < ovf time

No of Count = $\frac{\text{Req}}{\text{ovf}} = \frac{1}{0.78}$



Preload = $\text{ovf} \times (1 - 0.4)$ 256 μsec
 $256 \times (1 - 0.78125)$ 200 μsec

Normal Mode law's

- ① clk time = $\frac{\text{Prescaler}}{\text{system freq}}$
- ② ovf time = ovf value × clk time → $2^n \times \text{clk time}$
- ③ # of ovf Count = $\frac{\text{Req time}}{\text{ovf time}}$
- ④ Pre load value = ovf value × (1 - 0.4)
 $2^n (1 - 0.4)$

ex: timer lubit fcpu = 8 MHz Prescaler = 8

Req time = 5 sec

clk time = $\frac{\text{Prescaler}}{\text{Fcpu}} = \frac{8}{8\text{MHz}} = 1\text{ }\mu\text{sec}$

ovf time = $2^n \times \text{clk time} = 1024 \times 1\text{ }\mu\text{sec} \Rightarrow 1024\text{ }\mu\text{sec.}$

No of Count = $\frac{\text{Req}}{\text{ovf}} = \frac{5000000}{1024} = 4882.8125$

Preload = $2^n \times (1 - 0.4) = 1024 \times (1 - 0.8125) = 192$

* Code

void Init() {
 timer Reg = 192;
 Normal mode
 En Interrupt (ovf)
}

ISR(ovf) {
 static uint16_t Count = 0;
 Count++;
 if (Count == 4883)
 {
 Action
 Count = 0;
 timer Reg = 192;
 }
}

timer → Compare match

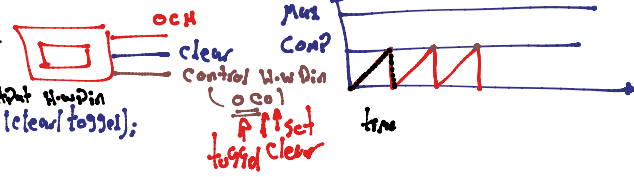
* Compare Match Mode →

① Clear timer compare match (CTC)

① set oc flag

② clear timer register

③ Control output window
↳ (set/clear/toggle);



CTC law's

① $\text{Clk time} = \frac{\text{Prescaler}}{\text{system freq}}$

② Compare time = Compare value * Clk time.

③ # of CTC Count = $\frac{\text{Req time Compare match time}}{\text{Compare time}}$

of CTC Count = $\frac{\text{Req time}}{\text{Compare value * clk time}}$

* Assume ab CTC Count to get the compare value.

↳ ① must be Integer

↳ ② must be cover value in Range.

ex: toggle led each $\boxed{1000 \mu\text{sec}}$, $\boxed{1000 \text{ msec}}$

timer = 8 bit $f_{\text{CPU}} = 8 \mu\text{Hz}$ Prescaler = 8

$\text{Clk time} = \frac{8}{8 \mu} = 1 \mu\text{sec}$

No of CTC Count = $\frac{\text{Req time}}{\text{Compare value * Clk time}}$

$\boxed{\text{Compare value}} * \text{Clk time} = \frac{\text{Req time}}{\text{No of CTC Count}}$

Compare value = $\frac{1000}{4} = \boxed{250}$